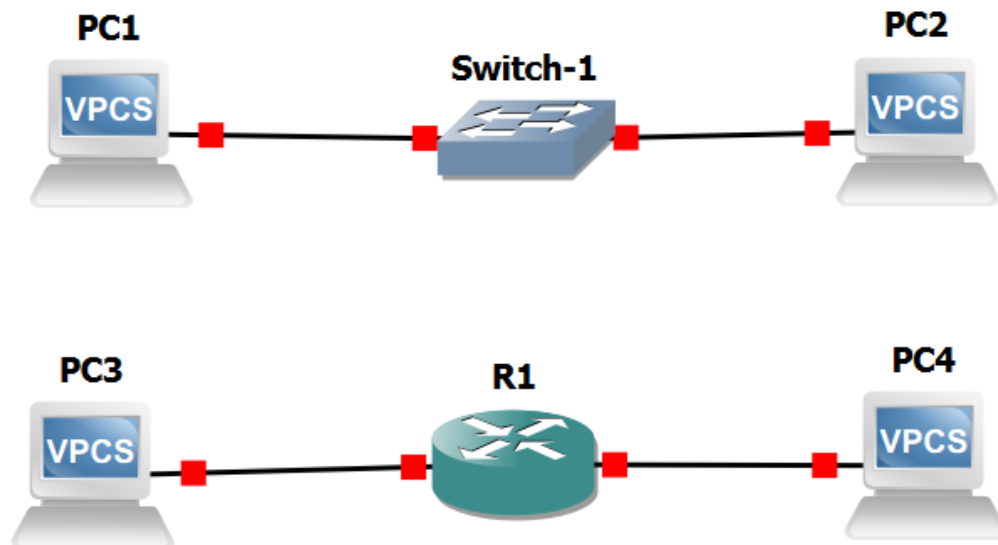


Лабораторная работа №1



Тема: Освоение инструментария для выполнения работ, построение простой сети

Все команды для настройки включаются в отчет в текстовом виде, не скриншоты.

nb! - отметка в тексте, "обратите особое внимание"

1) Установить и настроить эмулятор GNS3

2) Создать простейшую сеть, состоящую из 1 коммутатора и 2 компьютеров, назначить им произвольные ip адреса из одной сети

```
PC1> ip 192.168.1.2/24
```

Checking for duplicate address...

```
PC1 : 192.168.1.2 255.255.255.0
```

```
PC2> ip 192.168.1.3/24
```

Checking for duplicate address...

```
PC2 : 192.168.1.3 255.255.255.0
```

3) Запустить симуляцию, выполнить команду ping с одного из компьютеров, используя ip адрес второго компьютера

```
PC1> ping 192.168.1.3
```

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=6.631 ms

84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=8.670 ms

84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=0.811 ms

84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=5.572 ms

84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=0.691 ms

4) Перехватить трафик протокола arp на всех линках(nb!), задокументировать и проанализировать заголовки пакетов в программе Wireshark, для фильтрации трафика, относящегося к указанному протоколу использовать фильтры Wireshark

No.	Time	Source	Destination	Protocol	Length	Info
12	15.405531	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.3? Tell 192.168.1.2
13	15.405626	Private_66:68:01	Private_66:68:00	ARP	64	192.168.1.3 is at 00:50:79:66:68:01

<

>

> Frame 12: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▼ Address Resolution Protocol (request)
 Hardware type: Ethernet (1)
 Protocol type: IPv4 (0x0800)
 Hardware size: 6
 Protocol size: 4
 Opcode: request (1)
 Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)
 Sender IP address: 192.168.1.2
 Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
 Target IP address: 192.168.1.3

0000

ff ff ff ff ff ff 00 50 79 66 68 00 08 06 00 01

.....P yfh.....

0010

08 00 06 04 00 01 00 50 79 66 68 00 c0 a8 01 02

.....P yfh.....

0020

ff ff ff ff ff ff c0 a8 01 03 00 00 00 00 00 00

.....P yfh.....

0030

00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

.....

No.	Time	Source	Destination	Protocol	Length	Info
12	15.405531	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.3? Tell 192.168.1.2
13	15.405626	Private_66:68:01	Private_66:68:00	ARP	64	192.168.1.3 is at 00:50:79:66:68:01

<

>

> Frame 13: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface 0
> Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)
▼ Address Resolution Protocol (reply)
 Hardware type: Ethernet (1)
 Protocol type: IPv4 (0x0800)
 Hardware size: 6
 Protocol size: 4
 Opcode: reply (2)
 Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)
 Sender IP address: 192.168.1.3
 Target MAC address: Private_66:68:00 (00:50:79:66:68:00)
 Target IP address: 192.168.1.2

0000

00 50 79 66 68 00 00 50 79 66 68 01 08 06 00 01

Pyfh...P yfh.....

0010

08 00 06 04 00 02 00 50 79 66 68 01 c0 a8 01 03

.....P yfh.....

0020

00 50 79 66 68 00 c0 a8 01 02 00 00 00 00 00 00

Pyfh...P yfh.....

0030

00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

.....

5) Создать простейшую сеть, состоящую из 1 маршрутизатора и 2 компьютеров, назначить им произвольные ip адреса из разных сетей

PC3> ip 192.168.1.5/24 192.168.1.1

Checking for duplicate address...

PC3 : 192.168.1.5 255.255.255.0 gateway 192.168.1.1

PC3> save

Saving startup configuration to startup.vpc

. done

PC4> ip 192.168.2.4/24 192.168.2.1

Checking for duplicate address...

PC4 : 192.168.2.4 255.255.255.0 gateway 192.168.2.1

PC4> save

Saving startup configuration to startup.vpc

. done

6) Запустить симуляцию, выполнить команду ping с одного из компьютеров, используя IP-адрес второго компьютера

R1#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R1(config)#interface fastethernet 0/0

R1(config-if)#ip address 192.168.1.1 255.255.255.0

R1(config-if)#no shutdown

R1(config-if)#exit

R1(config)#interface fastethernet 1/0

R1(config-if)#ip address 192.168.2.1 255.255.255.0

R1(config-if)#no shutdown

R1(config-if)#

R1(config-if)#end

R1#write memory

Building configuration...

[OK]

PC3> ping 192.168.2.4

84 bytes from 192.168.2.4 icmp_seq=1 ttl=63 time=29.557 ms

84 bytes from 192.168.2.4 icmp_seq=2 ttl=63 time=15.808 ms

84 bytes from 192.168.2.4 icmp_seq=3 ttl=63 time=16.106 ms

84 bytes from 192.168.2.4 icmp_seq=4 ttl=63 time=14.864 ms

84 bytes from 192.168.2.4 icmp_seq=5 ttl=63 time=15.891 ms

7) Перехватить трафик протокола arp и icmp на всех линках(nb!), задокументировать и проанализировать заголовки пакетов в программе Wireshark, для фильтрации трафика, относящегося к указанному протоколу использовать фильтры Wireshark

icmp or arp							
No.	Time	Source	Destination	Protocol	Length	Info	
26	129.582056	Private_66:68:02	Broadcast	ARP	64	Who has 192.168.1.1? Tell 192.168.1.5	
27	129.588572	cc:01:19:e0:00:00	Private_66:68:02	ARP	60	192.168.1.1 is at cc:01:19:e0:00:00	
28	129.589494	192.168.1.5	192.168.2.4	ICMP	98	Echo (ping) request id=0xa1e2, seq=1/256, ttl=64 (reply in 29)	
29	129.618965	192.168.2.4	192.168.1.5	ICMP	98	Echo (ping) reply id=0xa1e2, seq=1/256, ttl=63 (request in 28)	
30	130.619407	192.168.1.5	192.168.2.4	ICMP	98	Echo (ping) request id=0xa2e2, seq=2/512, ttl=64 (reply in 31)	
31	130.635137	192.168.2.4	192.168.1.5	ICMP	98	Echo (ping) reply id=0xa2e2, seq=2/512, ttl=63 (request in 30)	
32	131.635391	192.168.1.5	192.168.2.4	ICMP	98	Echo (ping) request id=0xa3e2, seq=3/768, ttl=64 (reply in 33)	
33	131.651393	192.168.2.4	192.168.1.5	ICMP	98	Echo (ping) reply id=0xa3e2, seq=3/768, ttl=63 (request in 32)	
34	132.652470	192.168.1.5	192.168.2.4	ICMP	98	Echo (ping) request id=0xa4e2, seq=4/1024, ttl=64 (reply in 35)	
35	132.667245	192.168.2.4	192.168.1.5	ICMP	98	Echo (ping) reply id=0xa4e2, seq=4/1024, ttl=63 (request in 34)	
36	133.667438	192.168.1.5	192.168.2.4	ICMP	98	Echo (ping) request id=0xa5e2, seq=5/1280, ttl=64 (reply in 38)	
38	133.683250	192.168.2.4	192.168.1.5	ICMP	98	Echo (ping) reply id=0xa5e2, seq=5/1280, ttl=63 (request in 36)	

> Frame 26: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface 0

> Ethernet II, Src: Private_66:68:02 (00:50:79:66:68:02), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

> Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: Private_66:68:02 (00:50:79:66:68:02)

Sender IP address: 192.168.1.5

Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)

Target IP address: 192.168.1.1

0000 ff ff ff ff ff ff 00 50 79 66 68 02 08 06 00 01P yfh.....
0010 08 00 06 04 00 01 00 50 79 66 68 02 c0 a8 01 05P yfh.....
0020 ff ff ff ff ff ff c0 a8 01 01 00 00 00 00 00 00
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

icmp or arp							
No.	Time	Source	Destination	Protocol	Length	Info	
26	129.582056	Private_66:68:02	Broadcast	ARP	64	Who has 192.168.1.1? Tell 192.168.1.5	
27	129.588572	cc:01:19:e0:00:00	Private_66:68:02	ARP	60	192.168.1.1 is at cc:01:19:e0:00:00	
28	129.589494	192.168.1.5	192.168.2.4	ICMP	98	Echo (ping) request id=0xa1e2, seq=1/256, ttl=64 (reply in 29)	
29	129.618965	192.168.2.4	192.168.1.5	ICMP	98	Echo (ping) reply id=0xa1e2, seq=1/256, ttl=63 (request in 28)	
30	130.619407	192.168.1.5	192.168.2.4	ICMP	98	Echo (ping) request id=0xa2e2, seq=2/512, ttl=64 (reply in 31)	
31	130.635137	192.168.2.4	192.168.1.5	ICMP	98	Echo (ping) reply id=0xa2e2, seq=2/512, ttl=63 (request in 30)	
32	131.635391	192.168.1.5	192.168.2.4	ICMP	98	Echo (ping) request id=0xa3e2, seq=3/768, ttl=64 (reply in 33)	
33	131.651393	192.168.2.4	192.168.1.5	ICMP	98	Echo (ping) reply id=0xa3e2, seq=3/768, ttl=63 (request in 32)	
34	132.652470	192.168.1.5	192.168.2.4	ICMP	98	Echo (ping) request id=0xa4e2, seq=4/1024, ttl=64 (reply in 35)	
35	132.667245	192.168.2.4	192.168.1.5	ICMP	98	Echo (ping) reply id=0xa4e2, seq=4/1024, ttl=63 (request in 34)	
36	133.667438	192.168.1.5	192.168.2.4	ICMP	98	Echo (ping) request id=0xa5e2, seq=5/1280, ttl=64 (reply in 38)	
38	133.683250	192.168.2.4	192.168.1.5	ICMP	98	Echo (ping) reply id=0xa5e2, seq=5/1280, ttl=63 (request in 36)	

> Frame 27: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0

> Ethernet II, Src: cc:01:19:e0:00:00 (cc:01:19:e0:00:00), Dst: Private_66:68:02 (00:50:79:66:68:02)

> Address Resolution Protocol (reply)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: reply (2)

Sender MAC address: cc:01:19:e0:00:00 (cc:01:19:e0:00:00)

Sender IP address: 192.168.1.1

Target MAC address: Private_66:68:02 (00:50:79:66:68:02)

Target IP address: 192.168.1.5

0000 00 50 79 66 68 02 cc 01 19 e0 00 00 08 06 00 01 ..Pyfh.....
0010 08 00 06 04 00 02 cc 01 19 e0 00 00 c0 a8 01 01 ..Pyfh.....
0020 00 50 79 66 68 02 c0 a8 01 05 00 00 00 00 00 00 ..
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

icmp or arp						
No.	Time	Source	Destination	Protocol	Length	Info
26	129.582056	Private_66:68:02	Broadcast	ARP	64	Who has 192.168.1.1? Tell 192.168.1.5
27	129.588572	cc:01:19:e0:00:00	Private_66:68:02	ARP	60	192.168.1.1 is at cc:01:19:e0:00:00
→ 28	129.589494	192.168.1.5	192.168.2.4	ICMP	98	Echo (ping) request id=0xa1e2, seq=1/256, ttl=64 (reply in 29)
← 29	129.618965	192.168.2.4	192.168.1.5	ICMP	98	Echo (ping) reply id=0xa1e2, seq=1/256, ttl=63 (request in 28)
→ 30	130.619407	192.168.1.5	192.168.2.4	ICMP	98	Echo (ping) request id=0xa2e2, seq=2/512, ttl=64 (reply in 31)
← 31	130.635137	192.168.2.4	192.168.1.5	ICMP	98	Echo (ping) reply id=0xa2e2, seq=2/512, ttl=63 (request in 30)
→ 32	131.635391	192.168.1.5	192.168.2.4	ICMP	98	Echo (ping) request id=0xa3e2, seq=3/768, ttl=64 (reply in 33)
← 33	131.651393	192.168.2.4	192.168.1.5	ICMP	98	Echo (ping) reply id=0xa3e2, seq=3/768, ttl=63 (request in 32)
→ 34	132.652470	192.168.1.5	192.168.2.4	ICMP	98	Echo (ping) request id=0xa4e2, seq=4/1024, ttl=64 (reply in 35)

> Frame 28: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bit	0000	cc 01 19 e0 00 00 00 50	79 66 68 02 08 00 45 00	P yfh...E
> Ethernet II, Src: Private_66:68:02 (00:50:79:66:68:02), Dst: cc:01:19:e0:	0010	00 54 e2 a1 00 00 40 01	13 ae c0 a8 01 05 c0 a8	.T....@.....
> Internet Protocol Version 4, Src: 192.168.1.5, Dst: 192.168.2.4	0020	02 04 08 00 7e 28 a1 e2	00 01 08 09 0a 0b 0c 0d	(.....
0100 = Version: 4	0030	0e 0f 10 11 12 13 14 15	16 17 18 19 1a 1b 1c 1d	
.... 0101 = Header Length: 20 bytes (5)	0040	1e 1f 20 21 22 23 24 25	26 27 28 29 2a 2b 2c 2d	..!"#\$\$% &'()*+,-
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)	0050	2e 2f 30 31 32 33 34 35	36 37 38 39 3a 3b 3c 3d	./012345 6789;,<=
Total Length: 84	0060	3e 3f		>?
Identification: 0xe2a1 (58017)				
> 000. = Flags: 0x0				
...0 0000 0000 0000 = Fragment Offset: 0				
Time to Live: 64				
Protocol: ICMP (1)				
Header Checksum: 0x13ae [validation disabled]				
[Header checksum status: Unverified]				
Source Address: 192.168.1.5				
Destination Address: 192.168.2.4				
[Stream index: 0]				
> Internet Control Message Protocol				

icmp or arp						
No.	Time	Source	Destination	Protocol	Length	Info
26	129.582056	Private_66:68:02	Broadcast	ARP	64	Who has 192.168.1.1? Tell 192.168.1.5
27	129.588572	cc:01:19:e0:00:00	Private_66:68:02	ARP	60	192.168.1.1 is at cc:01:19:e0:00:00
→ 28	129.589494	192.168.1.5	192.168.2.4	ICMP	98	Echo (ping) request id=0xa1e2, seq=1/256, ttl=64 (reply in 29)
← 29	129.618965	192.168.2.4	192.168.1.5	ICMP	98	Echo (ping) reply id=0xa1e2, seq=1/256, ttl=63 (request in 28)
→ 30	130.619407	192.168.1.5	192.168.2.4	ICMP	98	Echo (ping) request id=0xa2e2, seq=2/512, ttl=64 (reply in 31)
← 31	130.635137	192.168.2.4	192.168.1.5	ICMP	98	Echo (ping) reply id=0xa2e2, seq=2/512, ttl=63 (request in 30)
→ 32	131.635391	192.168.1.5	192.168.2.4	ICMP	98	Echo (ping) request id=0xa3e2, seq=3/768, ttl=64 (reply in 33)
← 33	131.651393	192.168.2.4	192.168.1.5	ICMP	98	Echo (ping) reply id=0xa3e2, seq=3/768, ttl=63 (request in 32)
→ 34	132.652470	192.168.1.5	192.168.2.4	ICMP	98	Echo (ping) request id=0xa4e2, seq=4/1024, ttl=64 (reply in 35)

> Frame 29: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bit	0000	00 50 79 66 68 02 cc 01	19 e0 00 00 08 00 45 00	..Pyfh... ..E
> Ethernet II, Src: cc:01:19:e0:00:00 (cc:01:19:e0:00:00), Dst: Private_66:	0010	00 54 e2 a1 00 00 3f 01	14 ae c0 a8 02 04 c0 a8	.T....?.....
> Internet Protocol Version 4, Src: 192.168.2.4, Dst: 192.168.1.5	0020	01 05 00 00 86 28 a1 e2	00 01 08 09 0a 0b 0c 0d	(.....
0100 = Version: 4	0030	0e 0f 10 11 12 13 14 15	16 17 18 19 1a 1b 1c 1d	
.... 0101 = Header Length: 20 bytes (5)	0040	1e 1f 20 21 22 23 24 25	26 27 28 29 2a 2b 2c 2d	..!"#\$\$% &'()*+,-
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)	0050	2e 2f 30 31 32 33 34 35	36 37 38 39 3a 3b 3c 3d	./012345 6789;,<=
Total Length: 84	0060	3e 3f		>?
Identification: 0xe2a1 (58017)				
> 000. = Flags: 0x0				
...0 0000 0000 0000 = Fragment Offset: 0				
Time to Live: 63				
Protocol: ICMP (1)				
Header Checksum: 0x14ae [validation disabled]				
[Header checksum status: Unverified]				
Source Address: 192.168.2.4				
Destination Address: 192.168.1.5				
[Stream index: 0]				
> Internet Control Message Protocol				